Group Theory

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Example 1. Let $GL(2,\mathbb{R})$[or, it is denoted as $GL_2(\mathbb{R})$] be the multiplicative group of all real non-singular matrices of order 2. Show that the set

$$H = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2,\mathbb{R}) : det(A) = ad - bc = 1 \right\}$$

is a subgroup of $GL(2,\mathbb{R})$.

Solution. Since $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in H$, so H is a non-empty set.

Let $A, B \in H$. Then $\det A = 1$, $\det B = 1$.

As, $det(B) = 1(\neq 0)$, so B^{-1} exists and $det(B^{-1}) = \frac{1}{det(B)} = 1$, so $B^{-1} \in H$.

Now, as $A, B^{-1} \in H$, so by closure property, $AB^{-1} \in H$.

Therefore, H is a subgroup of $GL(2,\mathbb{R})$.

Example 2. The subset H of a group $(G, \circ)$ is defined by Centre

$$H = \{x \in G : x \circ g = g \circ x \quad \forall g \in G\}$$

Prove that H is a subgroup of G.

Solution. Since the identity element $e \in G$ and for all $g \in G$, $e \circ g = g \circ e$, so $e \in H$. Therefore H is a non-empty set.

Let $h_1, h_2 \in H$. To show H is a subgroup of G, we will show that $h_1 h_2^{-1} \in H$.

Then $h_1 \circ g = g \circ h_1$ and $h_2 \circ g = g \circ h_2 \quad \forall g \in G$.

First we will show, $h_2^{-1} \in H$.

Now, $h_2 \circ g = g \circ h_2 \quad \forall g \in G$, implies,

$h_2^{-1} \circ (h_2 \circ g) \circ h_2^{-1} = h_2^{-1} \circ (g \circ h_2) \circ h_2^{-1}$.

$\implies (h_2^{-1} \circ h_2) \circ (g \circ h_2^{-1}) = (h_2^{-1} \circ g) \circ (h_2 \circ h_2^{-1})$.

$\implies e \circ (g \circ h_1^{-1}) = (h_1^{-1} \circ g) \circ e \implies g \circ h_1^{-1} = h_1^{-1} \circ g$, which implies that $h_2^{-1} \in H$.

Now, $(h_1 \circ h_2^{-1}) \circ g = h_1 \circ (h_2^{-1} \circ g) = h_1 \circ (g \circ h_2^{-1}) = (h_1 \circ g) \circ h_2^{-1} = (g \circ h_1) \circ h_2^{-1} = g \circ (h_1 \circ h_2^{-1}) \quad \forall g \in G$.

So, $h_1 \circ h_2^{-1} \in H$ and hence H is a subgroup of G.

Note: The above subgroup H is called the centre of the group G, and is denoted by $Z(G)$. Clearly from definition, $Z(G)$ is an abelian group.

Example 3. Let G be an abelian group. Prove that the subset $S = \{p \in G : p = p^{-1}\}$ forms a subgroup of G.

Solution. Since $e = e^{-1}$, so $e \in S$. Therefore $S \neq \emptyset$.

Let $x, y \in S$. Then, $x = x^{-1}$ and $y = y^{-1}$. We will show $xy^{-1} \in S$.

As $y \in S$, so $y = y^{-1}$. $\therefore y^{-1} \in S$.

Now, as G is abelian, $(xy^{-1})^{-1} = x^{-1}y = xy^{-1}$, implies that $xy^{-1} \in S$.

Hence S is a subgroup of G.

Example 4. Consider the group $(\mathbb{Z}, +)$ and the subset $n\mathbb{Z} := \{\cdots, -3n, -2n, -n, 0, n, 2n, 3n, \cdots\}$. Show that, $n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.

Solution. Let, $x, y \in n\mathbb{Z}$. Then, $x = sn$ and $y = tn$ for some $s, t \in \mathbb{Z}$. Clearly 0 is the identity element of $(\mathbb{Z}, +)$. Since, $t, n \in \mathbb{Z}$, so, $y = tn \in \mathbb{Z}$. As, $\mathbb{Z}$ is a group, so inverse of y (denoted as $-y$) exists and it is $-tn$.

Now, $x + (-y) = sn - tn = (s - t)n \in n\mathbb{Z}$, as $s - t \in \mathbb{Z}$. Hence, $n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.

Theorem 5. *Let G be a finite cyclic group generated by x. Then $o(G) = n$ iff $o(x) = n$.*

Theorem 6. *A finite group G is cyclic iff there exists an element x such that $o(x) = o(G)$.*

Theorem 7. *If $G = \langle x \rangle$ be a cyclic group of order n then for any integer r (where, $1 \leq r < n$), a^r is a generator of G iff $gcd(r, n) = 1$.*

Theorem 8. *n^{th} root of unity forms a cyclic group under multiplication. That is, the set of all solutions of the equation $x^n = 1$ forms a cyclic group under multiplication.*

Example 9. Show that the set $G = \{1, 2, 3, 4, 5, 6\}$ forms a cyclic group under multiplication modulo 7. Find all generators of this group.

Solution. First show that G forms a group under multiplication, by showing four properties of a group. Clearly it is shown that, $o(G) = 6$ and 1 is the identity element of G under multiplication. We have,

$$\begin{aligned}
3^1 &= 3 \\
3^2 &= 9 = 2 \text{ in modulo } 7 \\
3^3 &= 27 = 6 \text{ in modulo } 7 \\
3^4 &= 81 = 4 \text{ in modulo } 7 \\
3^5 &= 243 = 5 \text{ in modulo } 7 \\
3^6 &= 729 = 1 \text{ in modulo } 7
\end{aligned}$$

So, 3 is a generator of G and hence G is cyclic.

Otherwise, it is seen that, $o(3) = 6 = o(G)$, so by Theorem 6, we can conclude that G is cyclic.

Similarly, it can be seen that, 5 is also a generator of the group G.

Example 10. Show that the group $(\mathbb{Z}_5, +)$ is cyclic. Find all of its generators.

Solution. Here the set $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$, or simply $\{0, 1, 2, 3, 4\}$. Clearly, under the addition, 0 is the identity element in $\mathbb{Z}_5$. Since, $1+1+1+1+1 = 5 = 0$ in $\mathbb{Z}_5$, so $o(1) = 5 = o(\mathbb{Z}_5)$. Then, by Theorem 6, $\mathbb{Z}_5$ is a cyclic group under addition. Now, check 2 is also a generator:

$$\begin{aligned}
2 + 2 &= 4 \\
2 + 2 + 2 &= 6 = 1 \text{ in modulo } 5 \\
2 + 2 + 2 + 2 &= 8 = 3 \text{ in modulo } 5 \\
2 + 2 + 2 + 2 + 2 &= 10 = 0 \text{ in modulo } 5
\end{aligned}$$

Similarly, it is easy to check that, 3, 4 are also generators of $\mathbb{Z}_5$.

Note 11. If n is a prime then, $(\mathbb{Z}_n, +)$ forms a cyclic group and all of its elements except identity are generators. That is, $1, 2, \cdots, n-1$ are all generators of $\mathbb{Z}_n$.

Example 12. Let G be a cyclic group of order 40 and generated by a, i.e., $G = \langle a \rangle$. Consider the cyclic subgroups $\langle a^8 \rangle$ and $\langle a^{10} \rangle$. Find all distict elements of these two subgroups.

Solution. Given $o(G) = 40$ and $G = \langle a \rangle$, so $o(a) = 40$, implies $a^{40} = e$, the identity of G.

Now, $(a^8)^5 = a^{40} = e$, so $o(a^8) = 5$ and therefore,

$$\begin{aligned}\langle a^8 \rangle &= \{(a^8)^0, (a^8)^1, (a^8)^2, (a^8)^3, (a^8)^4\} \\ &= \{e, a^8, a^{16}, a^{24}, a^{32}\}.\end{aligned}$$

Similarly, $\langle a^{10} \rangle = \{e, a^{10}, a^{20}, a^{30}\}$.

Theorem 13 (Lagrange's Theorem)**.** *The order of any subgroup of a finite group divides order of the group.*

Proof. Let H be a subgroup of a finite group G and $o(G) = n$, $o(H) = m$. Consider the set of all distinct left cosets of H in G. Since, G contains finite number of elements, so the number of distinct left cosets of H in G is finite. Let, $\{x_1H, x_2H, \cdots, x_rH\}$ be the collection of all distinct left cosets of H in G. Therefore, $G = x_1H \cup x_2H \cup \cdots \cup x_rH$. Since, $o(H) = m$, so each left cosets x_iH contains exactly m-elements. That is, for each $i = 1, 2, \cdots, r$, $o(x_iH) = m$. Then,

$$\begin{aligned}&o(G) = o(x_1H) + o(x_2H) + \cdots + o(x_rH) \\ &\Rightarrow n = m + m + \cdots + m(r\text{-times}) = mr,\end{aligned}$$

which implies, $m|n$. □

Note 14. 1. $[G : H] = \dfrac{o(G)}{o(H)}$.

2. If G is a group of prime order p, then G can have no proper subgroups.

3. *The converse of the Lagrange's Theorem is not true.* For example, the alternating group A_4 is of order 12 but there is no subgroup of A_4 of order 6, though 6 is a divisor of 12.

Theorem 15. *Every group of prime order is cyclic.*

Proof. Let G be a group of prime order p. Since p is a prime, $o(G) > 1$ and hence G has an element $a \neq e$. Let H be a cyclic subgroup generated by a. That is $H = \langle a \rangle$. Then $o(H) > 1$. By Lagrange's theorem $o(H)|o(G)$. As p is prime, so $o(H) = p = o(G)$, implies $H = G$. Hence, G is cyclic. □

Definition 16. A subgroup H of a group G is said to be normal if $aH = Ha$ $\forall a \in G$. Then in symbol, we write $H \triangle G$.

Note 17. 1. $aH = Ha$ does not imply that $ah = ha$ $\forall h \in H$. But from the definition it follows that every subgroup of a commutative group is a normal subgroup.

2. Every group G has at least two normal subgroups, namely G itself and $\{e\}$. These are called improper or trivial normal subgroups. All others are called proper normal subgroups.

3. For normal subgroups, left cosets and right cosets are same.

Definition 18. A group having no proper normal subgroups is called a Simple Group.

Note 19. By Lagrange's Theorem, every group of prime order has no proper subgroups. Hence every group of prime order is simple.

Theorem 20. *Let H be a subgroup of a group G. Then H is normal in G if and only if $xhx^{-1} \in H \ \forall h \in H$ and $\forall x \in G$.*

Proof. **(if part)** Let H be a normal subgroup of G.
Then $xH = Hx \ \forall x \in G$.
Let $x \in G$ and $h \in H$. Then, $xh \in xH = Hx$, so $xh = h_1x$ for some $h_1 \in H$, which implies $xhx^{-1} = h_1 \Rightarrow xhx^{-1} \in H$, as $h_1 \in H$.
(Only if part) Conversely, let $xhx^{-1} \in H \ \forall h \in H$ and $\forall x \in G$.
Let $a \in Hx$. $a = h_2x$ for some $h_2 \in H$. Then,
$a = (xx^{-1})h_2x = x(x^{-1}h_2x) = xh_3$, where $h_3 = x^{-1}h_2(x^{-1})^{-1} \in H$, as $x^{-1} \in G$.
Thus, $a \in Hx \Rightarrow a \in xH$, as $xh_3 \in xH$. So,$Hx \subseteq xH$ ... (i)
Next, let $b \in xH$. So ,$b = xh_4$ for some $h_4 \in H$.
Then, $b = xh_4(x^{-1}x) = (xh_4x^{-1})x = h_5x$, where $h_5 = xh_4x^{-1} \in H$.
So, $b \in xH \Rightarrow b \in Hx$, as $h_5x \in Hx$. So, $xH \subseteq Hx$ ... (ii)
Then from (i) and (ii), we have, $xH = Hx \ \forall x \in G$
Hence H is normal in G. □

Theorem 21. *The intersection of any two normal subgroups of a group is a normal subgroup.*

Proof. Let H_1 and H_2 be any two normal subgroups of a group G. Then $H_1 \cap H_2$ is a subgroup of G as H_1 and H_2 are subgroups of G. Let x be any element of G and $h \in H_1 \cap H_2$. Then $h \in H_1$ and $h \in H_2$.
so, $xhx^{-1} \in H_1$, as H_1 is a normal subgroup of G;
and $xhx^{-1} \in H_2$, as H_2 is a normal subgroup of G.
So,$xhx^{-1} \in H_1 \cap H_2$.
Thus, $x \in G$ and $h \in H_1 \cap H_2 \Rightarrow xhx^{-1} \in H_1 \cap H_2$
Hence $H_1 \cap H_2$ is a normal subgroup of G. □

Note 22.
1. The intersection of any collection of normal subgroups of a group is a normal subgroup of G.

2. Union of two normal subgroups need not be a normal subgroup. For example, consider the group $(\mathbb{Z}, +)$ and its two subgroups $2\mathbb{Z}$ and $3\mathbb{Z}$. Since $(\mathbb{Z}, +)$ is abelian group, so all of its subgroups are normal and so $2\mathbb{Z}$ and $3\mathbb{Z}$ are normal in $\mathbb{Z}$. But $2\mathbb{Z} \cup 3\mathbb{Z}$ is not even a subgroup of $\mathbb{Z}$, as $2+3 = 5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$.

Example 23. Show that every subgroup of a cyclic group is normal.
Solution. Let G be a cyclic group and H be a subgroup of G.
Let $x \in G$ and $h \in H$. As G is cyclic, it is abelian.
Then $xhx^{-1} = xx^{-1}h = eh = h \in H$.
So, $x \in G, h \in H \Rightarrow xhx^{-1} \in H$, conclude that, H is a normal subgroup of G.

Definition 24. Let G be a group and H be its subgroup. Index of the subgroup H is denoted as $[G : H]$ and is defined as the number of distinct left(/right) cosets of H in G.

Example 25. If G is a group and H is a subgroup of index 2 in G, prove that H is a normal subgroup of G.
Solution. Since $[G : H] = 2$, so there are only two distinct left (/right) cosets of H in G and the two distinct left cosets are $H, G \setminus H$. Also, the two distinct right cosets are $H, G \setminus H$.

Let $x \in H$. Then $xH = H$ and $Hx = H$ so, $xH = Hx$.
Next let $x \notin H$. Then $x \in G \setminus H$
So, $xH = G \setminus H$, as $G \setminus H$ is the only left coset.
Also, $Hx = G \setminus H$ as $G \setminus H$ is the only right coset.
Thus, $xH = Hx \ \forall x \in G$
Hence, $H \triangle G$.

Example 26. Show that the centre of a group G, given by

$$Z(G) = \{a \in G : ag = ga \ \forall g \in G\}$$

is a normal subgroup of G.

Solution. It is already proved that $Z(G)$ is a subgroup of G.
Now for any $g \in G$ and any $a \in Z(G)$,
$gag^{-1} = agg^{-1} = ae = a \in Z(G)$.
Hence $Z(G)$ is a normal subgroup of G.

Example 27. Prove that the set of matrices

$$H = \left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} : x \in \mathbb{R}, x \neq 0 \right\}$$

forms a normal subgroup of $GL(2, \mathbb{R})$, the group of all real non-singular 2×2 matrices.

Solution. Let $A = \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \in H$, $B = \begin{pmatrix} y & 0 \\ 0 & y \end{pmatrix} \in H$
Then $x \in \mathbb{R}, x \neq 0$ and $y \in \mathbb{R}, y \neq 0$
Now $|B| = y^2 \neq 0$ so, B^{-1} exists.

$B^{-1} = \frac{1}{y^2} \begin{pmatrix} y & 0 \\ 0 & y \end{pmatrix} = \begin{pmatrix} \frac{1}{y} & 0 \\ 0 & \frac{1}{y} \end{pmatrix}$

$AB^{-1} = \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \begin{pmatrix} \frac{1}{y} & 0 \\ 0 & \frac{1}{y} \end{pmatrix} = \begin{pmatrix} \frac{x}{y} & 0 \\ 0 & \frac{x}{y} \end{pmatrix} \in H$, as $\frac{x}{y} \neq 0$ and $\frac{x}{y} \in \mathbb{R}$

so, $A, B \in H \Rightarrow AB^{-1} \in H$ so, H is a subgroup of $GL(2, \mathbb{R})$.

Next let $C = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix} \in G$, whose determinant, $|C| \neq 0$

$C^{-1} = \frac{1}{x_1 y_2 - x_2 y_1} \begin{pmatrix} y_2 & -y_1 \\ -x_2 & x_1 \end{pmatrix}$

$CAC^{-1} = \frac{1}{x_1 y_2 - x_2 y_1} \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix} \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \begin{pmatrix} y_2 & -y_1 \\ -x_2 & x_1 \end{pmatrix}$

$= \frac{1}{x_1 y_2 - x_2 y_1} \begin{pmatrix} x_1 x & y_1 x \\ x_2 x & y_2 x \end{pmatrix} \begin{pmatrix} y_2 & -y_1 \\ -x_2 & x_1 \end{pmatrix}$

$= \frac{1}{x_1 y_2 - x_2 y_1} \begin{pmatrix} x_1 x y_2 - y_1 x x_2 & -x_1 x y_1 + y_1 x x_1 \\ x_2 x y_2 - y_2 x x_2 & -x_2 x y_1 + y_2 x x_1 \end{pmatrix}$

$= \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \in H$

$\therefore A \in H, C \in G \Rightarrow CAC^{-1} \in H$
$\therefore H$ is a normal subgroup of $GL(2, \mathbb{R})$.

Example 28. Let H is a subgroup of G, and define

$$N(H) := \{g \in G : gHg^{-1} = H\}.$$

Show that,

1. $N(H)$ is a subgroup of G.
2. H is a normal subgroup of $N(H)$.

Solution.

1. Let $a, b \in N(H)$. Then $aHa^{-1} = H$, $bHb^{-1} = H$ Now $bHb^{-1} = H \implies b^{-1}Hb = H$. Then, $(ab)H(ab)^{-1} = abH(b^{-1}a^{-1}) = a(bHb^{-1})a^{-1} = aHa^{-1} = H$ Hence $ab^{-1} \in N(H)$. Thus $a, b \in N(H) \implies ab^{-1} \in N(H)$

   So, $N(H)$ is a subgroup of G.

2. Next let $h \in H$. Then $hHh^{-1} = H$, as H is a subgroup. Hence $h \in N(H)$.

   Therefore, $H \subseteq N(H)$

   Now $N(H)$ is a subgroup of G, H is a subgroup of G and $H \subseteq N(H)$. Therefore, $H = N(H)$ and so H is a subgroup of $N(H)$.

   Let $p \in N(H)$. Then $pHp^{-1} = H \implies pHp^{-1} \in N(H)$. Therefore H is a normal subgroup of $N(H)$.

Quotient Group: Let H be a normal subgroup of a group G. Then there is no distinction between left and right cosets of H in G. Let G/H be the collection of all cosets of H in G. We shall now show that the set G/H forms a group w.r.t multiplication of cosets defined by $(aH)(bH) = abH$, $\forall a, b \in G$.

Theorem 29. *The set G/H of all cosets of a normal subgroup H in G forms a group w.r.t operation defined by*

$$(aH)(bH) = abH, \quad \forall a, b \in G.$$

Proof. Here $G/H = \{aH : a \in G\}$.
Let $a, b, c \in G$ so that $aH, bH, cH \in G/H$.

(i) Then $(aH)(bH) = abH \in G/H$ as $a, b \in G \implies ab \in G$. So, G/H is closed under the given operation.

(ii) Now $(aH)[(bH)(cH)] = aH(bcH) = a(bc)H = (ab)cH = (ab)(cH) = [(aH)(bH)](cH)$. Thus the given operation is associative.

(iii) We have $H = eH \in G/H$. Then $(H)(aH) = (eH)(aH) = (ea)H = aH$ and similarly $(aH)(H) = aH$. Hence H is the identity element of G/H.

(iv) As $a \in G \implies a^{-1} \in G$, so $aH \in G/H \implies a^{-1}H \in G/H$.

Now $(aH)(a^{-1}H) = (aa^{-1})H = eH = H$ and $(a^{-1}H)(aH) = (a^{-1}a)H = eH = H$. Therefore, $(aH)^{-1} = a^{-1}H$. Thus inverse of each element of G/H exist. Hence G/H is a group.

□

Note 30. The group G/H is known as Quotient Group.

Theorem 31. *Every quotient group of an abelian group is abelian.*

Proof. Let H be a subgroup of an abelian group G. Then H is a normal subgroup of G. So the quotient group G/H exists.

Let $a, b \in G$. Then $aH, bH \in G/H$.

Now, $(aH)(bH) = abH = baH$ [as G is abelian, so $ab = ba$] $= (bH)(aH)$.

so, $(aH)(bH) = (bH)(aH), \forall aH, bH \in G/H$

Hence the quotient group G/H is abelian. □

Theorem 32. *Every quotient group of a cyclic group is cyclic.*

Proof. Let H be a subgroup of a cyclic group G. Then H is a normal subgroup of G, as G is abelian. So the quotient group G/H exist. Let a be the generator of G. Then $a^n \in G$ for some integer n. Therefore $a^nH \in G/H$.

Now, $a^nH = (a \cdot a \cdots a \ (n\text{-times}))H = (aH)(aH)\cdots(aH)\ (n\text{times}) = (aH)^n$

Hence G/H is a cyclic group and aH is a generator of it. □

Example 33. Let $Z(G)$ be the centre of a group G. If G/Z is cyclic, prove that G is abelian.

Solution. Since G/Z is cyclic, let $G/Z = \langle gZ \rangle$ for some $g \in G$.

Let $a \in G$. Then $aZ \in G/Z$.

Therefore $aZ = (gZ)^m$ for some integer m.

Now, $(gZ)^m = (gZ)(gZ)\cdots(gZ)(m\text{-times}) = (gg\cdots g \ (m\text{-times}))Z = g^mZ$.

So, $a \in g^mZ \Rightarrow a = g^mz_1$, some $z_1 \in Z$

Next let $b \in G$. Then $b = g^nz_2$, for some $z_2 \in Z$.

Now,

$$\begin{aligned} ab &= (g^mz_1)(g^nz_2) \\ &= g^mg^nz_1z_2 \ \text{ as, } z_1 \in Z \text{ it commutes with } g^n \\ &= g^{m+n}z_1z_2 \end{aligned}$$

and

$$\begin{aligned} ba &= (g^nz_2)(g^mz_1) \\ &= g^ng^mz_2z_1 \ \text{ as, } z_2 \in Z \text{ it commutes with } g^m \\ &= g^{n+m}z_1z_2 \\ &= g^{n+m}z_2z_1 \end{aligned}$$

Thus $ab = ba \quad \forall a, b \in G$. Hence G is abelian.

Example 34. If N is normal in G and $a \in G$, then prove that $o(Na)$ in G/N is a divisor of $o(a)$.

Solution. Let $o(a) = n$ and $o(Na) = m$

Then $a^n = e$, the identity element of G , so $Na^n = Ne = N$

Now $Na^n = N(a \cdot a \cdot a \cdots a \ (n\text{-times})) = (Na)(Na)\cdots(Na)\ (n\text{-times}) = (Na)^n$

Thus $(Na)^n = N$, the identity of G/N. But $o(Na) = m$, so m divides n.